



CIS3200 Term Project Tutorial



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Lab Tutorial

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Global Weather Pattern Analysis and Prediction Using the Elastic Stack and Machine Learning

Objectives

In this hands-on lab, you will learn how to:

- Define and apply a custom index mapping and upload data into Elasticsearch using Git Bash.
- Create Data Views in Kibana for initial data exploration.
- Verify and transform geographic data (latitude/longitude → geo_point) for map visualizations.
- Create various charts and map visualizations using Kibana's Analytics tools.
- Assemble all visualizations into a comprehensive Global Weather Dashboard.

- Upload the dataset into the Elastic Cloud using Git Bash.
- Build two machine learning models (Multi-Metric and Regression) using the dataset.

Platform Spec

Tool	Version/Details
Elasticsearch/Kibana	8.9.1
Git Bash	Latest Version
Dataset	GlobalWeatherRepository (from Kaggle)
File Format	Weather_bulk.json (~28.4 MB) Download from GitHub
System	LocalHost deployment (Windows/MAC)
OS	Windows 10/11

Cluster Specifications

Cluster Version	8.9.1
Number of Nodes	1
Memory Size (RAM)	~ 16 GB
CPU (Number of Cores)	16 available processors

GET /

```
{
  "name": "KSLAPTOP",
  "cluster_name": "elasticsearch",
  "cluster_uuid": "B
-IkJzbUTXOFLStT5zInIQ",
  "version": {
    "number": "8.9.1",
```

GET /_nodes/os?pretty

```
{
  "_nodes": {
    "total": 1,
    "successful": 1,
    "failed": 0
```

```
"ml.machine_memory":
  "16406278144",
```

```
"available_processors": 16,  
"allocated_processors": 16
```

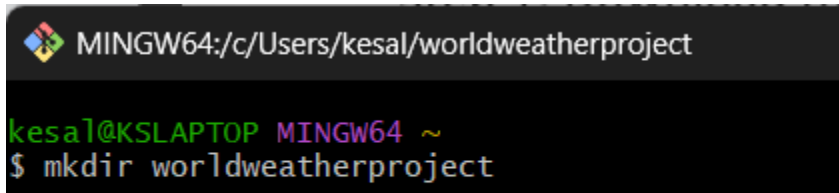
Step 1: Download the dataset to your local machine

This step involves manually downloading the JSON file from [GitHub](#), specifically the file named `weather_bulk.json`, due to Elasticsearch requires data in JSON or NDJSON format. The `weather_bulk.json` file is already preprocessed and structured correctly for bulk upload.

Once you have downloaded the dataset, we need to create a folder called 'worldweatherproject' using Git Bash.

1. Open Git Bash and use the command to create the folder:

```
mkdir worldweatherproject
```

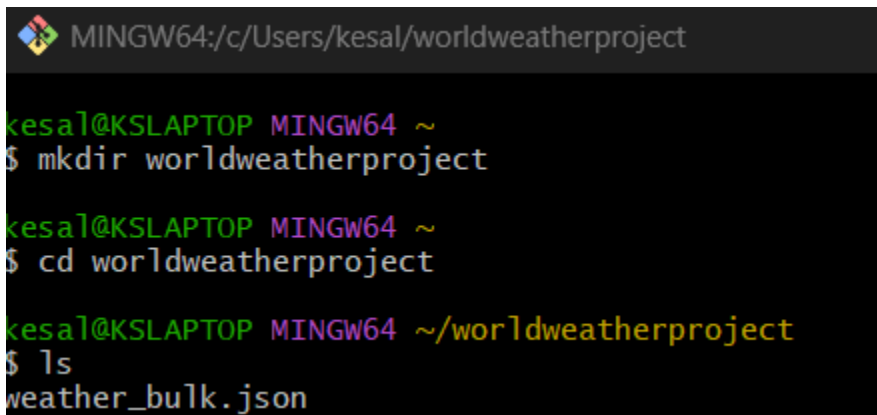


```
MINGW64:/c/Users/kesal/worldweatherproject  
  
kesal@KSLAPTOP MINGW64 ~  
$ mkdir worldweatherproject
```

2. Manually, drag the already downloaded [weather_bulk.json](#) file into the worldweatherproject folder.
3. Now, change your directory (folder) to 'worldweatherproject' from the current working directory using the '`cd`' command. Type in the command and hit the Enter key. Then, type in `ls` to list of folders and files in the current directory:

```
cd worldweatherproject
```

```
ls
```



```
MINGW64:/c/Users/kesal/worldweatherproject  
  
kesal@KSLAPTOP MINGW64 ~  
$ mkdir worldweatherproject  
  
kesal@KSLAPTOP MINGW64 ~  
$ cd worldweatherproject  
  
kesal@KSLAPTOP MINGW64 ~/worldweatherproject  
$ ls  
weather_bulk.json
```

4. Keep this window open for Step 4.

Step 2: Start Elasticsearch and Kibana

This step requires that both services (Elasticsearch and Kibana) be running before uploading the data. Elasticsearch is responsible for handling data indexing and queries, while Kibana provides exploration tools.

Note: Location name might be similar or vary between people

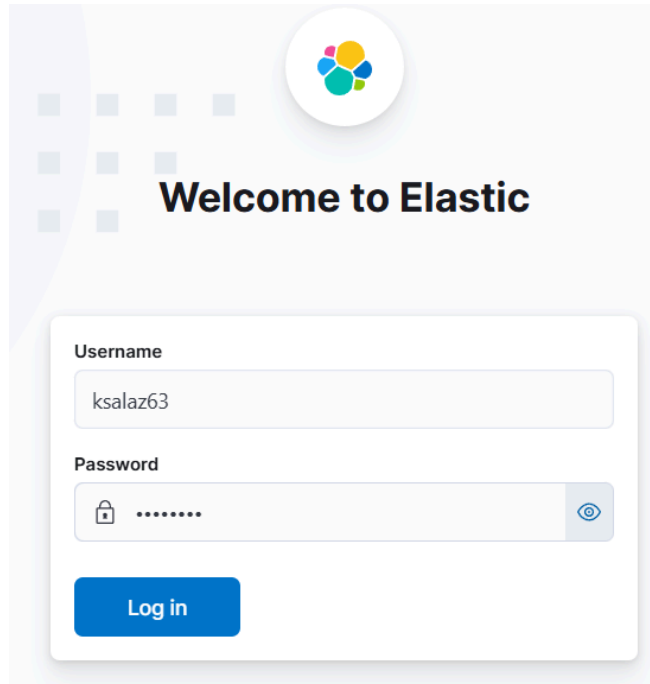
1. Open one CMD terminal on your PC. And, run the following command to start Elasticsearch from the command line. Type in the command “.bin\elasticsearch.bat” assuming your ES directory is located at “C:\Users\users\elasticsearch-8.9.1\bin\” as follows

```
C:\Users\kesal\elasticsearch-8.9.1\bin>elasticsearch.bat
```

2. You need to wait to see the message “current.health= “Green/Yellow” message” Cluster health status changed from [Red] to [Green/Yellow]” to see if Elasticsearch is running properly.
3. Open another terminal for Kibana. Start Kibana with the following command line “kibana.bat” at the location where it is located.

```
C:\Users\kesal\kibana-8.9.1\kibana-8.9.1\bin>kibana.bat
```

4. After a few minutes, you will receive the following message if Kibana runs successfully: “Kibana is now available (was degraded)”.
5. Open a web browser to the machine where you are running Kibana and specify the port number to enter the following URL to use Kibana: <http://localhost:5601>
6. Log in to your account using your username and password:



Step 3: Index Mapping Creation

This step creates an index using Git Bash called `weatherdata` and defines a mapping for its fields (data types). This defines how Elasticsearch should store and intercept each field (e.g., `geo_point`, date fields, and keyword fields). Without proper mapping, Elasticsearch may incorrectly guess data types, which can prevent accurate visualizations later.

1. Open a new Git Bash window, where we are going to use this command (change **username** and **password** with the one you use for ES) :

```
curl --cacert $ES_HOME/config/certs/http_ca.crt \
  -XPUT "https://username:password@localhost:9200/weatherdata" \
  -H 'Content-Type: application/json' \
  -d '{
    "mappings": {
      "properties": {
        "country": { "type": "keyword" },
        "location_name": { "type": "keyword" },
        "latitude": { "type": "float" },
        "longitude": { "type": "float" },
```

```

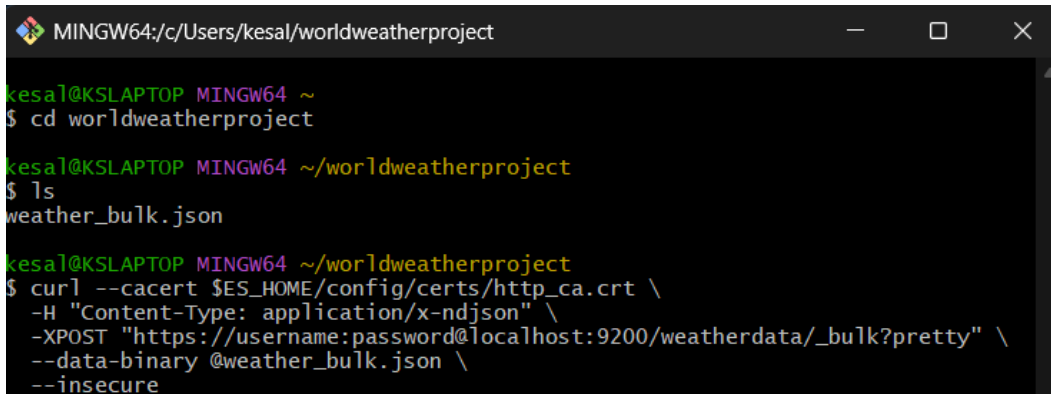
        "location": { "type": "geo_point" },
        "timezone": { "type": "keyword" },
        "last_updated": { "type": "date" },
        "temperature_celsius": { "type": "float" },
        "humidity": { "type": "integer" },
        "condition_text": { "type": "keyword" }
    }
}
}' \
--insecure

```

Step 4: Upload the JSON File to ElasticSearch

This step allows us to upload the weather dataset into the weatherdata index using the NDJSON file.

1. Using the first Git Bash window (used in Step 1), use this command (change **username** and **password** with the one you use for ES):



```

MINGW64:/c/Users/kesal/worldweatherproject
kesal@KSLAPTOP MINGW64 ~
$ cd worldweatherproject
kesal@KSLAPTOP MINGW64 ~/worldweatherproject
$ ls
weather_bulk.json
kesal@KSLAPTOP MINGW64 ~/worldweatherproject
$ curl --cacert $ES_HOME/config/certs/http_ca.crt \
-H "Content-Type: application/x-ndjson" \
-XPOST "https://username:password@localhost:9200/weatherdata/_bulk?pretty" \
--data-binary @weather_bulk.json \
--insecure

```

```

curl --cacert $ES_HOME/config/certs/http_ca.crt \
-H "Content-Type: application/x-ndjson" \
-XPOST
"https://username:password@localhost:9200/weatherdata/_bulk?pretty" \
--data-binary @weather_bulk.json \
--insecure

```

2. A successful upload will return a message showing:

```
"updated": 29640,
```

"failed": 0

Step 5: Create a Data View in Kibana

This step involves creating a Data View, which allows visualizations to reference the dataset. Kibana uses Data views to understand which Elasticsearch index it is working with, and the timestamp allows time-based filtering for dashboards and charts.

1. Go to **Kibana** → **Stack Management** → **Data Views**
2. Create a new Data View called wea*
3. The Timestamp field should be last_updated

Edit data view

Name

wea*

Index pattern

wea*

Timestamp field

last_updated

Select a timestamp field for use with the global time filter.

[Show advanced settings](#)

✓ Your index pattern matches 1 source.

All sources Matching sources

weatherdata Index

Rows per page: 10

4. Click Save.

Step 6: Preparing Geo-Fields and Standardizing Data

This step checks that Elasticsearch recognizes the location field as a geo_point and updates it accordingly. Geo_point fields are mandatory for creating maps, which is a visual we will be creating. These commands that we will be using will combine latitude and longitude into a single location coordinate recognized by Kibana's Maps feature. We will also be changing some locations to be in English, which were originally in a different language in the dataset.

1. Go to **Management** → **Dev Tools**

2. Run this command:

```
GET weatherdata/_mapping
```

3. Check if location is a geo_point:

```
{
  "location": {
    "type": "geo_point"
  }
}
```

4. Run this command to combine latitude and longitude into a single location coordinate:

```
POST weatherdata/_update_by_query
{
  "script": {
    "source": "ctx._source.location = ['lat': ctx._source.latitude,
'lon': ctx._source.longitude]"
  }
}
```

5. In the Dev tool, we will run these commands to change the Гватемала to Guatemala:

```
POST /weatherdata/_update_by_query?wait_for_completion=false
{
  "query": {
    "term": {
      "country": "Гватемала"
    }
  },
  "script": {
    "source": "ctx._source.country = 'Guatemala'",
    "lang": "painless"
  }
}
```

6. In the Dev tool, we will also run these commands to change Colombia to Colombia and update the location_name to Yopal.

```
POST /weatherdata/_update_by_query?wait_for_completion=false
{
  "query": {
    "term": {
      "country": "كولومبيا"
    }
  },
  "script": {
    "source": "ctx._source.country = 'Colombia'",
    "lang": "painless"
  }
}
```



```
}
```

```
POST /weatherdata/_update_by_query?wait_for_completion=false
{
  "query": {
    "term": {
      "country": "Colombia"
    }
  },
  "script": {
    "source": "ctx._source.location_name = 'Yopal'",
    "lang": "painless"
  }
}
```

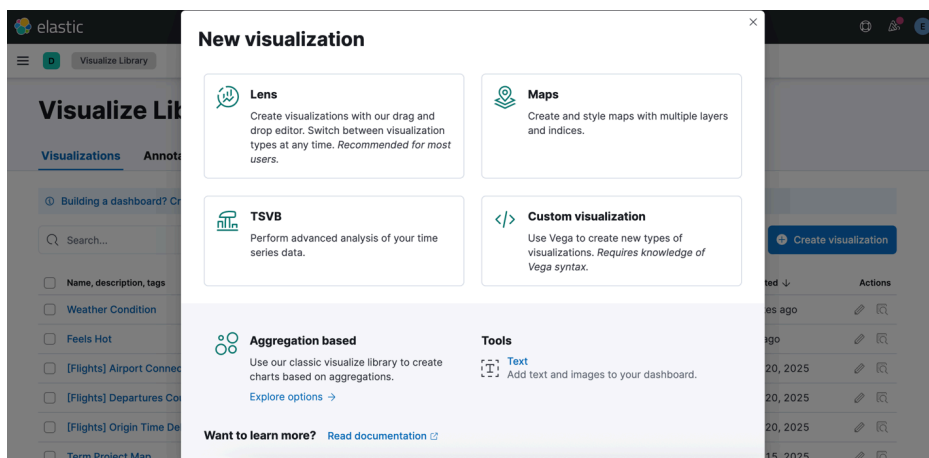
Step 7: Visualizations

This step will demonstrate how to create various visuals using the Wea* data view, which includes visuals such as Pie charts, Bar charts, and Maps.

***Note:** Any visual created will need to have a set date range. Choose Absolute with Start Date: May 10, 2024, at 18:00 and Absolute with End Date: October 31, 2024, at 23:30.*

Pie Chart- Global Distribution of Weather Conditions

1. From the Kibana main menu, **select Analytics → Visualize Library → Create Visualization**
2. Choose 'Aggregation Based', then select 'Pie Chart'.



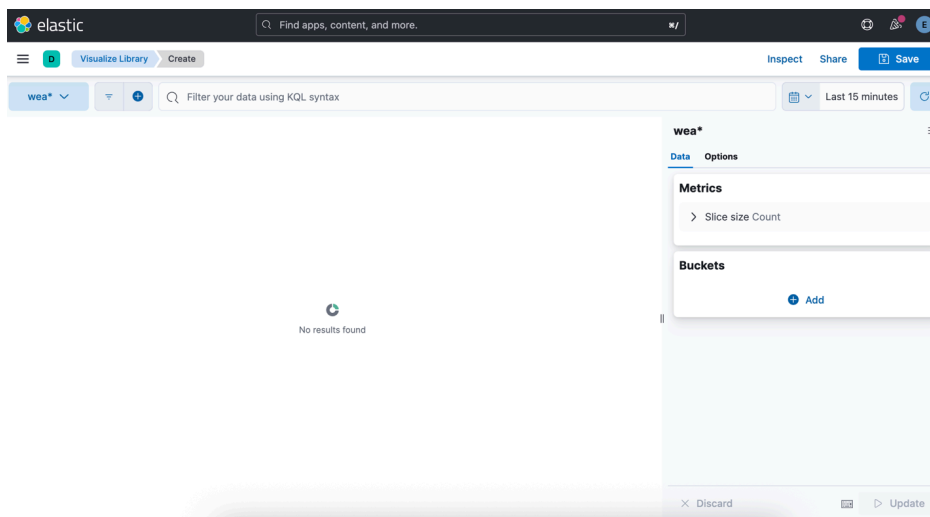
3. On the data source page, choose the newly created data view: wea*

New Pie / Choose a source

[Select a different visualization](#)

Type	Title ↑
	[Flights] Flight Log
	acc*
	com*
	Kibana Sample Data Flights
	logstash*
	Midterm
	shake*
	wea*

4. After choosing wea* as the source, the page will look like this:



5. When configuring the Pie Chart, in the Metrics section, set the Aggregation field to Count and type “Weather Condition” in the Custom Label field.

Metrics

✓ Slice size

Aggregation

Count [help](#) 

Count

Custom label

Weather Condition

- For the Buckets section, set the Aggregation field to Terms and set the Field to 'condition_text', while keeping the descending order and setting the size to 10.

Buckets

▼ Split slices  

Aggregation [Terms help](#) 

Terms ▼

Field

condition_text ▼

Order by

Metric: Weather Condition ▼

Order **Size**

Descending ▼ 10

- Click the time range in the top-right corner of Kibana. Select Absolute for Start and End Date, set Start Date as May 10, 2024, @ 18:00, and End Date as October 31, 2024, @ 23:30. Then click update to get the full visualization.

 ▼ May 10, 2024 @ 18:00:00.000 → Oct 31, 2024 @ 23:30:00.000

Absolute **Relative** **Now**

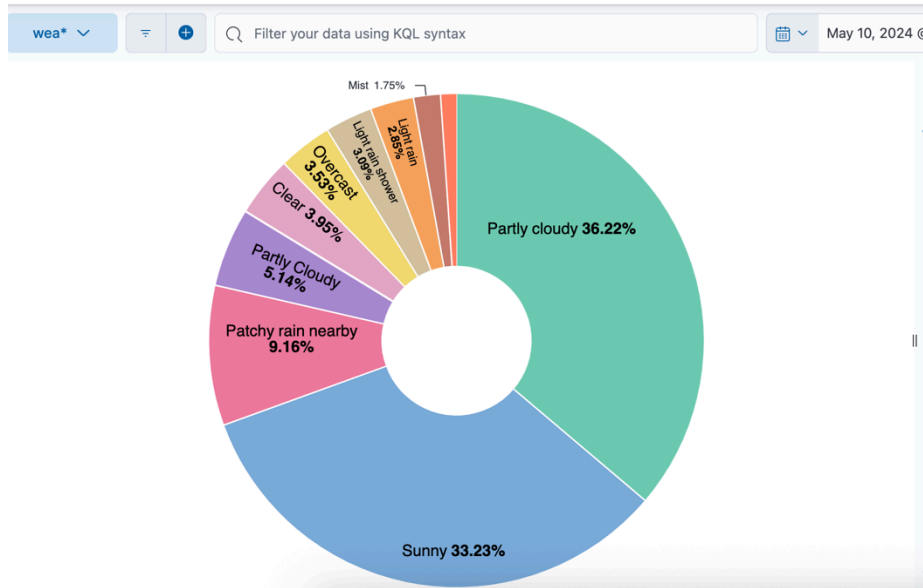
← **October** ▼ **2024** ▼ →

Su	Mo	Tu	We	Th	Fr	Sa	
							20:00
							20:30
29	30	1	2	3	4	5	21:00
6	7	8	9	10	11	12	21:30
13	14	15	16	17	18	19	22:00
20	21	22	23	24	25	26	22:30
27	28	29	30	31	1	2	23:00
							23:30

End date Oct 31, 2024 @ 23:30:00.000

- The results show a chart of which weather conditions dominate globally. It displays weather conditions such as Partly Cloudy, Sunny, Patchy rain nearby, Partly Cloudy, Clear, Overcast, Light

rain shower, Light rain, Mist, and Heavy Rain with Thunder. Partly Cloudy, Sunny, and Patchy rain nearby are the three dominant weather conditions.



9. Save the Pie chart, using the name 'Global Distribution of Weather Conditions' and under the section Add to Dashboard, clicking 'none'.

Save visualization

Title

Global Distribution of Weather Conditions

Description

Optional

Tags

Optional

Add to dashboard

☐ Existing

Search dashboards...

☐ New

☒ None

☒ Add to library ⓘ

☐ Save as new visualization

Cancel

Save

Pie Chart- Top 10 Cities Experiencing the Most Cumulative Heat

1. Repeat steps 1-4 from the previous visualization, keeping in mind to set the Time Filter.
2. When configuring the Pie Chart, in the Metrics section, set the Aggregation field to Sum, set Field to feels_like_celsius, and type 'Feels Like (°C)' in the Custom Label field.

Metrics

▼ Slice size

Aggregation

[Sum help](#)

Sum

Field

feels_like_celsius

Custom label

Feels Like (C*)

> Advanced

- For the Buckets section, set the Aggregation field to Terms and set the Field to 'location_name', while keeping the descending order and setting the size to 10.

Buckets

▼ Split slices



Aggregation

[Terms help](#)

Terms

Field

location_name

Order by

Metric: Feels Like (C*)

Order

Descending

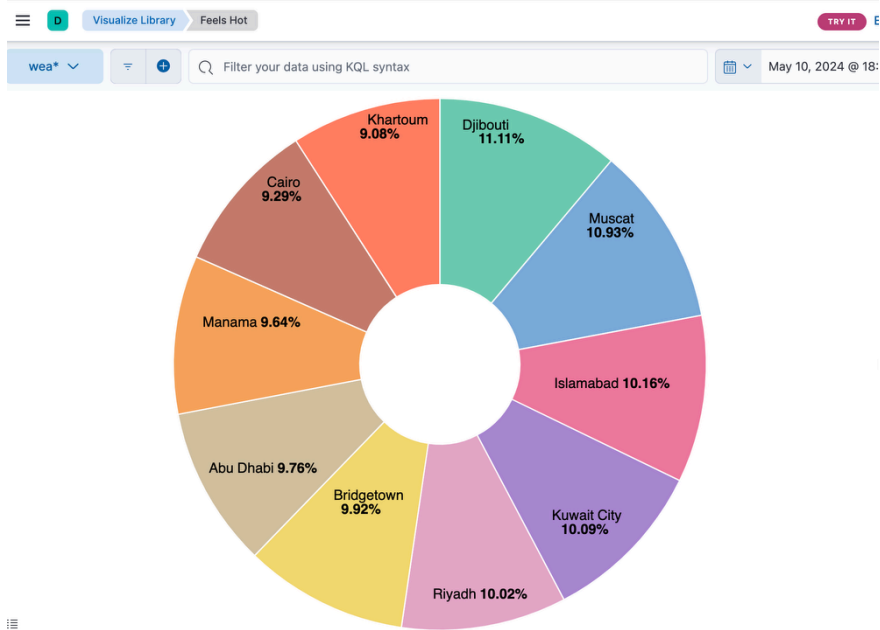
Size

10

☐ Group other values in separate bucket

☐ Show missing values

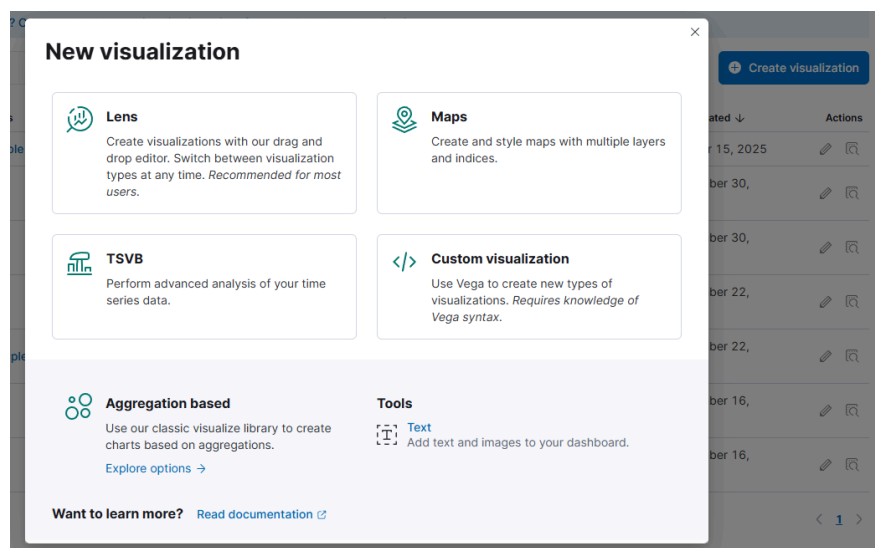
- With the Time filter set, click "Update" to view the full visualization. It'll show the cities that tend to feel the hottest are Khartoum, Djibouti, Muscat, Islamabad, Kuwait City, Riyadh, Bridgetown, Abu Dhabi, Manama, and Cairo.



5. Save the Pie chart, using the name 'Top 10 Cities Experiencing the Most Cumulative Heat,' and under the section Add to Dashboard, clicking 'none'.

Bar Chart- Top 10 Cities with the Highest Humidity

1. From the Kibana main menu, select Analytics → Visualize Library → Create Visualization
2. Choose 'Aggregation Based', then select 'Vertical Bar'.



3. On the data source page, choose the data view: wea*

New Vertical bar / Choose a source

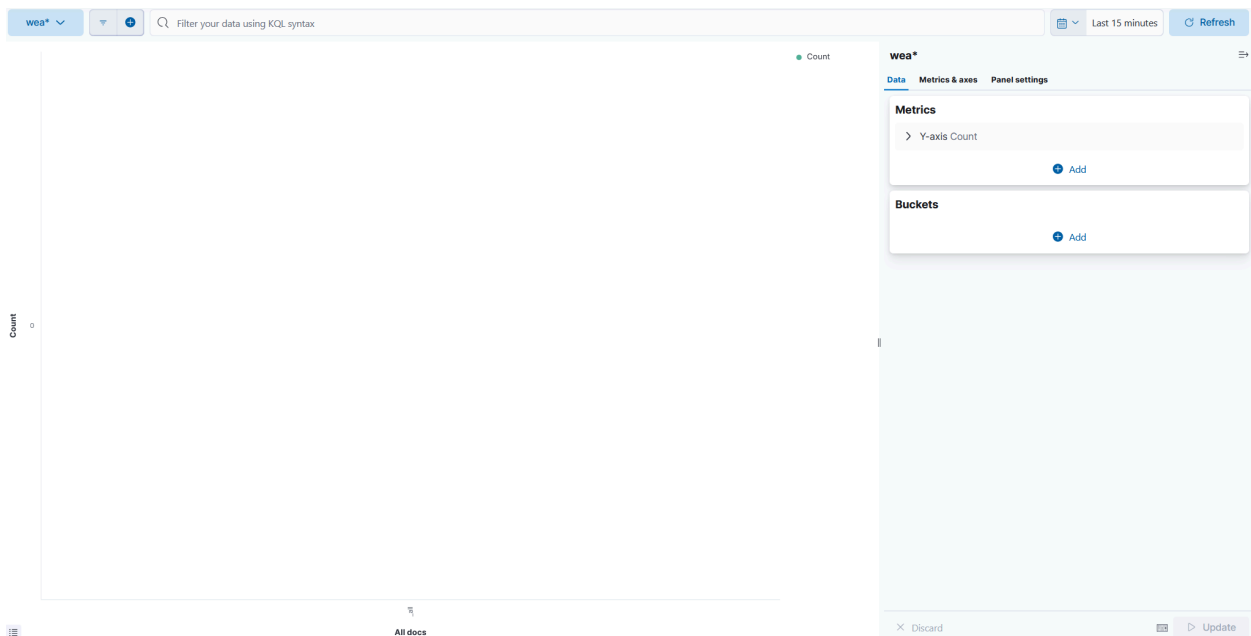
[Select a different visualization](#)

Types ▾

Type	Title ↑
	acc*
	com*
	logstash*
	shakespeare*
	wea*

< 1 >

- After choosing wea* as the source, a blank chart will appear with Metrics (Y-Axis) and Buckets (X-Axis).



- When configuring the Pie Chart, in the Metrics section, set the Aggregation field to "Average", the Field to "humidity", and type "Average Humidity" in the Custom Label field.

wea*

[Data](#) [Metrics & axes](#) [Panel settings](#)

Metrics

Y-axis

Aggregation [Average help](#)

Average

Field

humidity

Custom label

Average Humidity

> Advanced

+ Add

- For Buckets, click "Add" and select "X-Axis". Choose "Terms" for aggregation, "location_name" for the field, and type "Cities" in the Custom label field. For the order section, keep the descending option and put 10 for the size.

Buckets

X-axis

Aggregation [Terms help](#)

Terms

Field

location_name

Order by

Metric: Average Humidity

Order

Descending

Size

10

☐ Group other values in separate bucket

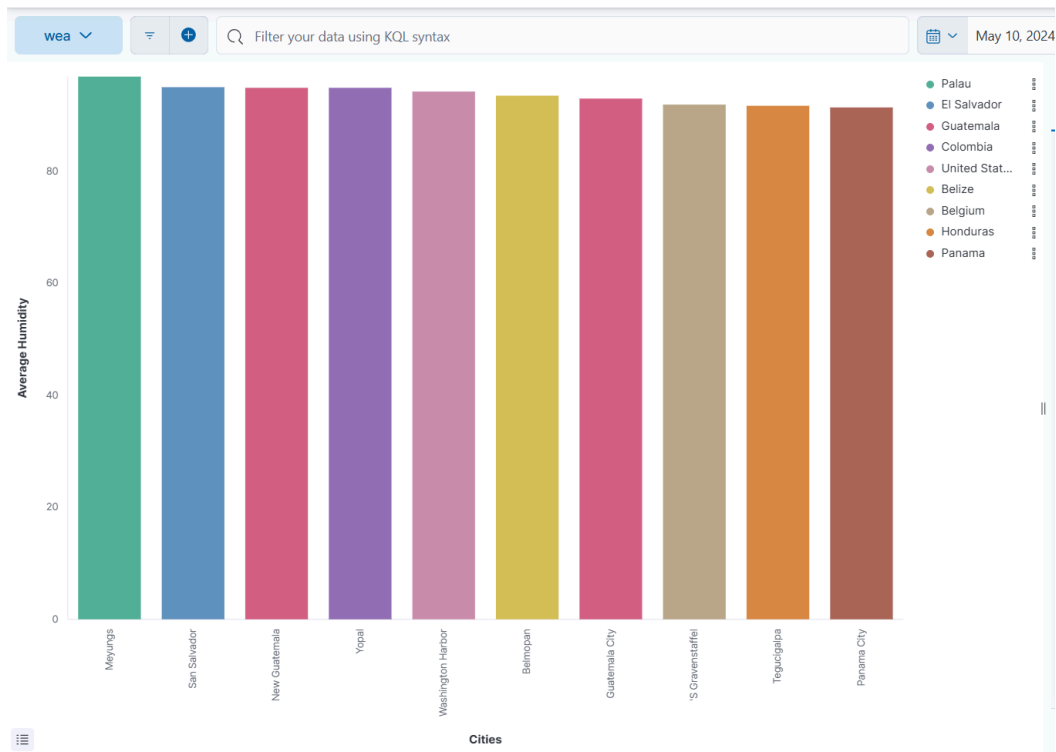
☐ Show missing values

Custom label

Cities

- In the Bucket section, click "+ Add" and choose "Split Series." Choose "Terms" for aggregation, "country" for the field. Keep the descending option and set the size to 10.

- The visualization will show the top 10 cities with the highest average humidity, including Meyungs, San Salvador, New Guatemala, Yopal, Washington Harbor, Belmopan, Guatemala City, S Gravenstaffel, Tegucigalpa, and Panama City.



- Save the Bar chart, using the name 'Top 10 Cities with the Highest Humidity' and under the section Add to Dashboard, clicking 'none'.

Bar Chart- Top 10 Countries with the Highest Average UV Index

- Repeat steps 1-3 from the previous bar chart, keeping in mind to set the Time Filter.
- When configuring the Pie Chart, in the Metrics section, set the Aggregation field to "Average", the Field to "uv_index", and type "Average UV Index" in the Custom Label field.

wea*

Data Metrics & axes Panel settings

Metrics

Y-axis

Aggregation [Average help](#)

Average

Field

uv_index

Custom label

Average UV Index

> Advanced

+ Add

3. For Buckets, click "Add" and select "X-Axis". Choose "Terms" for aggregation and "country" for the field. For the order section, keep the descending option and put 10 for the size. Type "Countries" in the Custom label field

Buckets

▼ X-axis

Aggregation

Terms

Field

country

Order by

Metric: Average UV Index

Order

Descending

Size

10

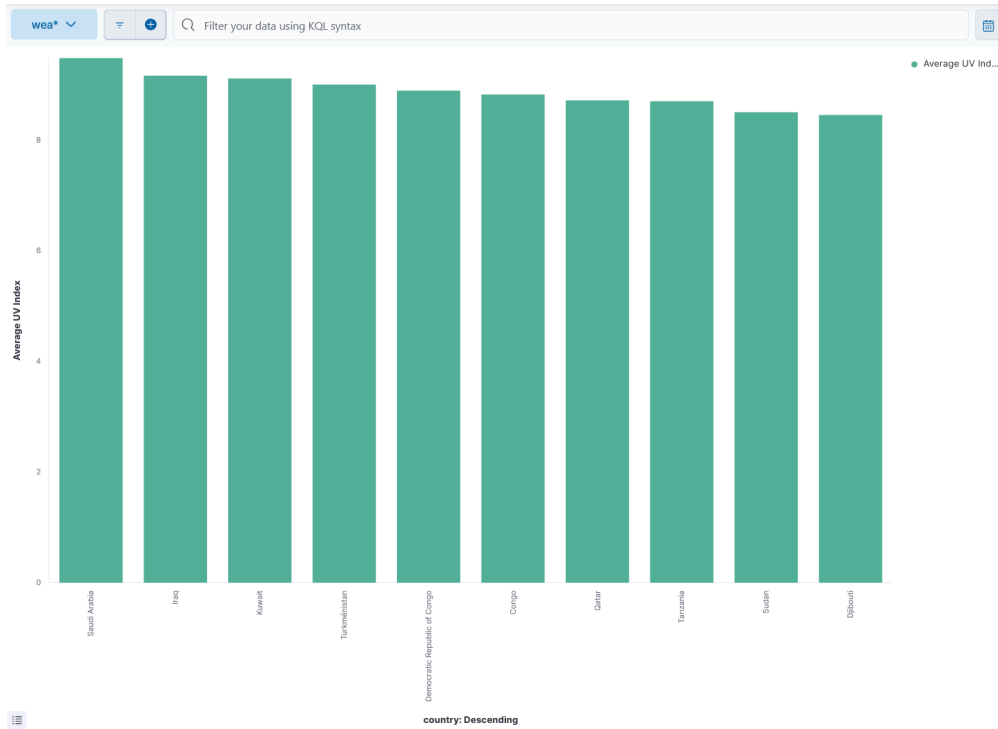
☐ Group other values in separate bucket

☐ Show missing values

Custom label

Countries

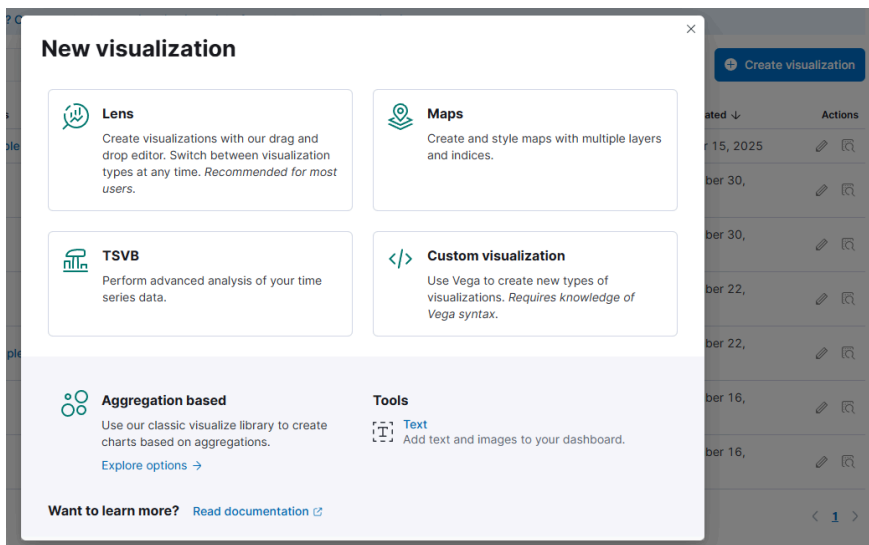
4. After you click "Update," a visualization of the 10 countries with the highest UV index will be displayed. Saudi Arabia, Iraq, Kuwait, Turkmenistan, Democratic Republic of Congo, Congo, Qatar, Tanzania, Sudan, and Djibouti.



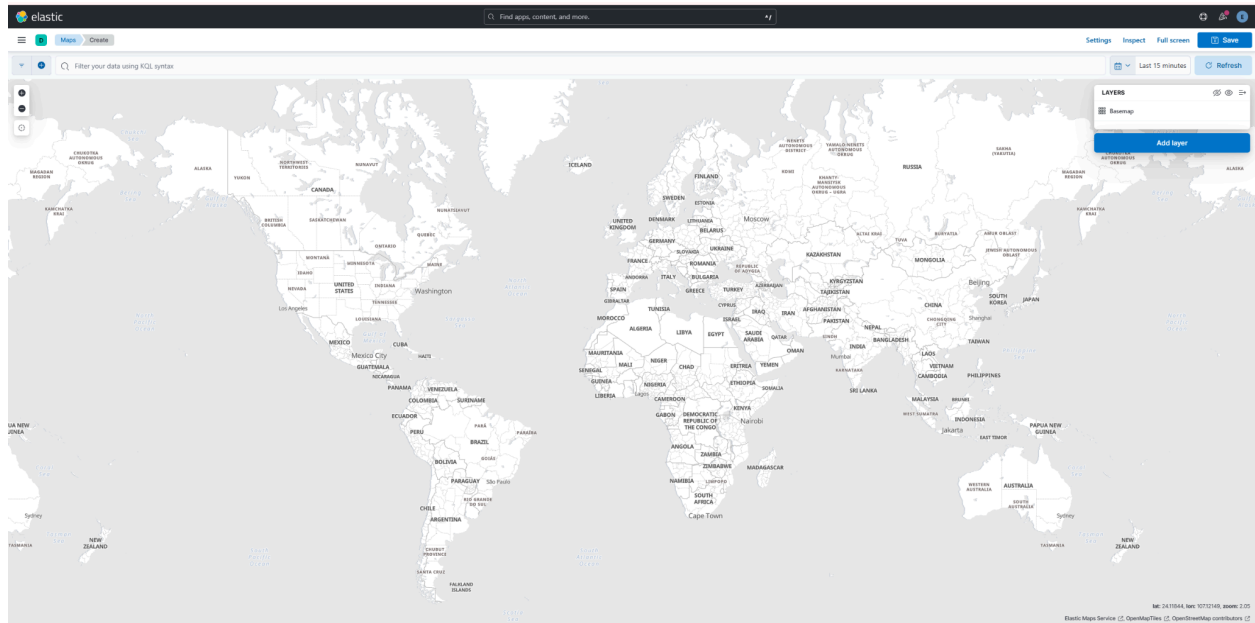
5. Save the Bar chart, using the name 'Top 10 Countries with the Highest Average UV Index' and under the section Add to Dashboard, clicking 'none'.

Map- UV Index Intensity by City

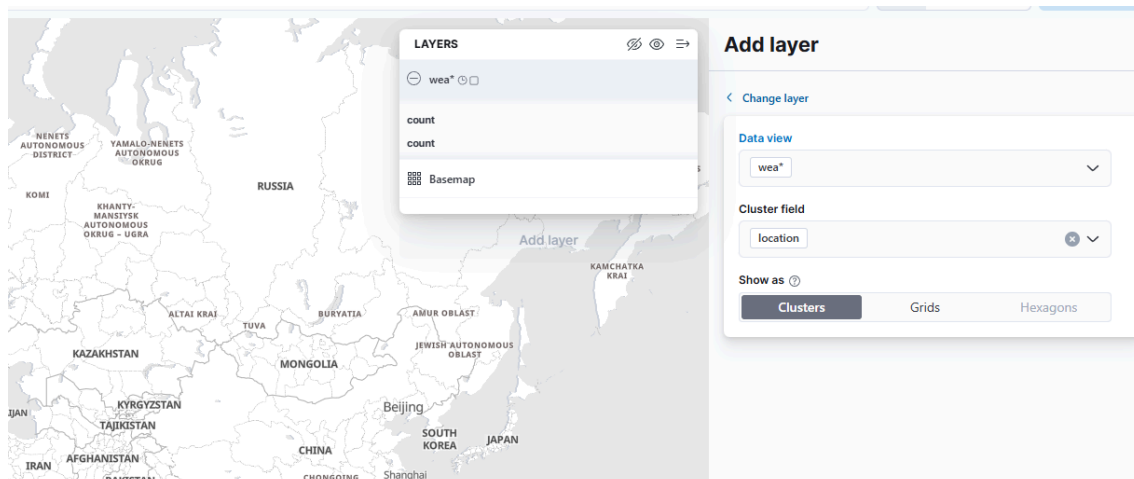
1. From the Kibana main menu, select Analytics → Visualize Library → Create Visualization
 - a. On the pop up, select Maps



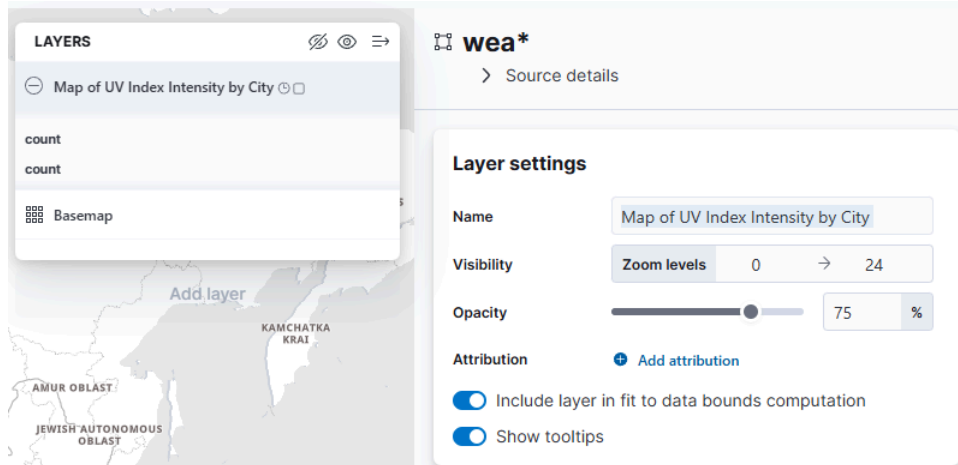
2. After selecting Maps, a blank map will appear. On the Map → Add Layer → Select Clusters



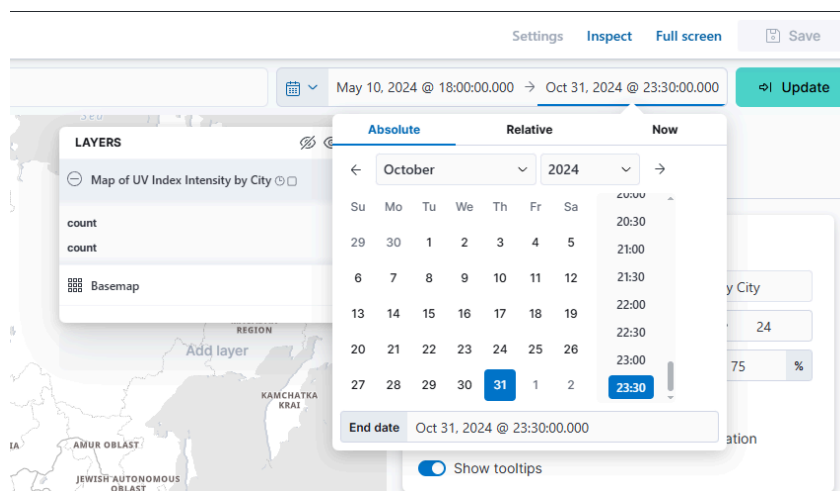
3. For the Data View, select `wea*` and click add and continue.



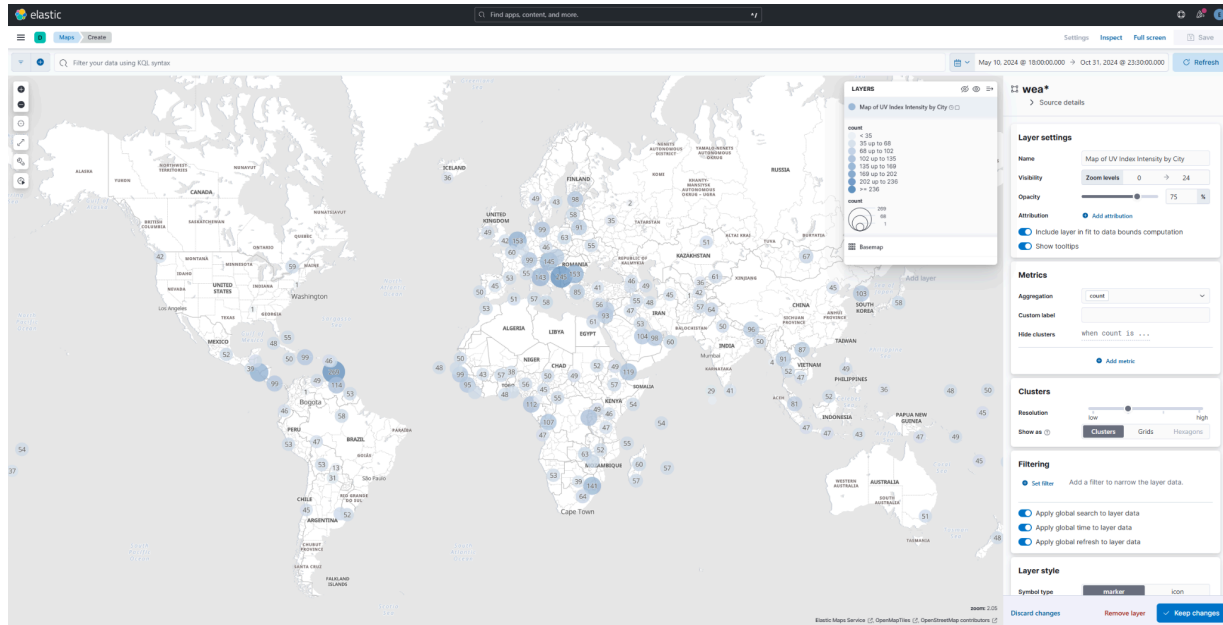
4. In the Layer settings, type “Map of UV Index Intensity by City” in the Name field.



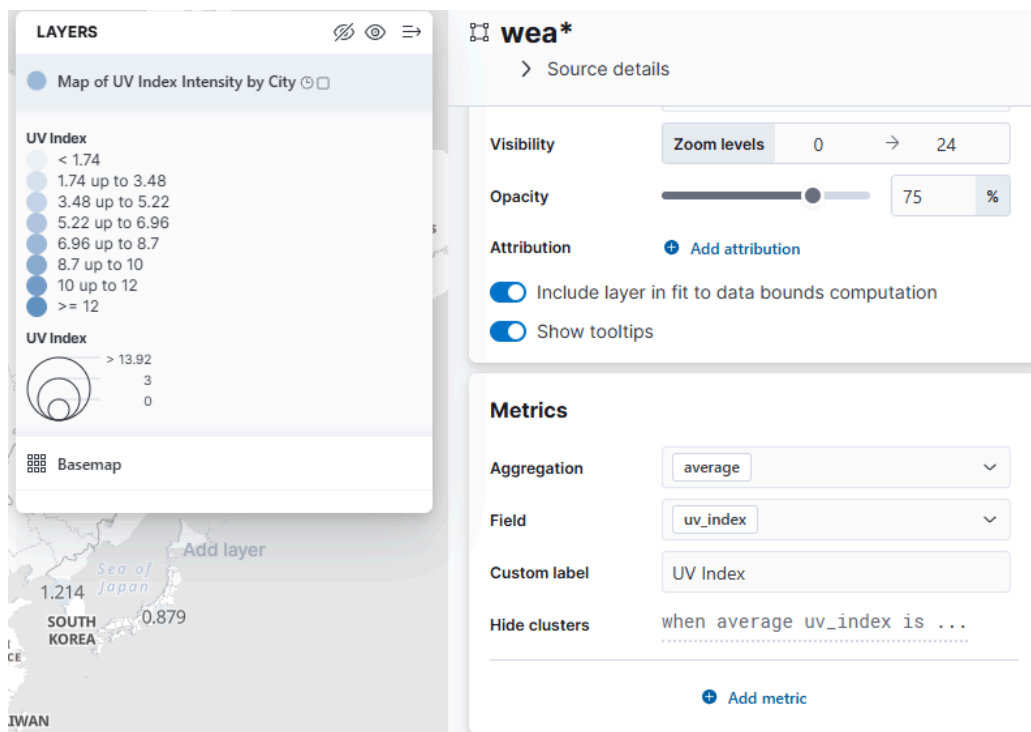
- Click the time range in the top-right corner of Kibana. Select Absolute for Start and End Date, set Start Date as May 10, 2024, @ 18:00, and End Date as October 31, 2024, @ 23:30. Then, click update.



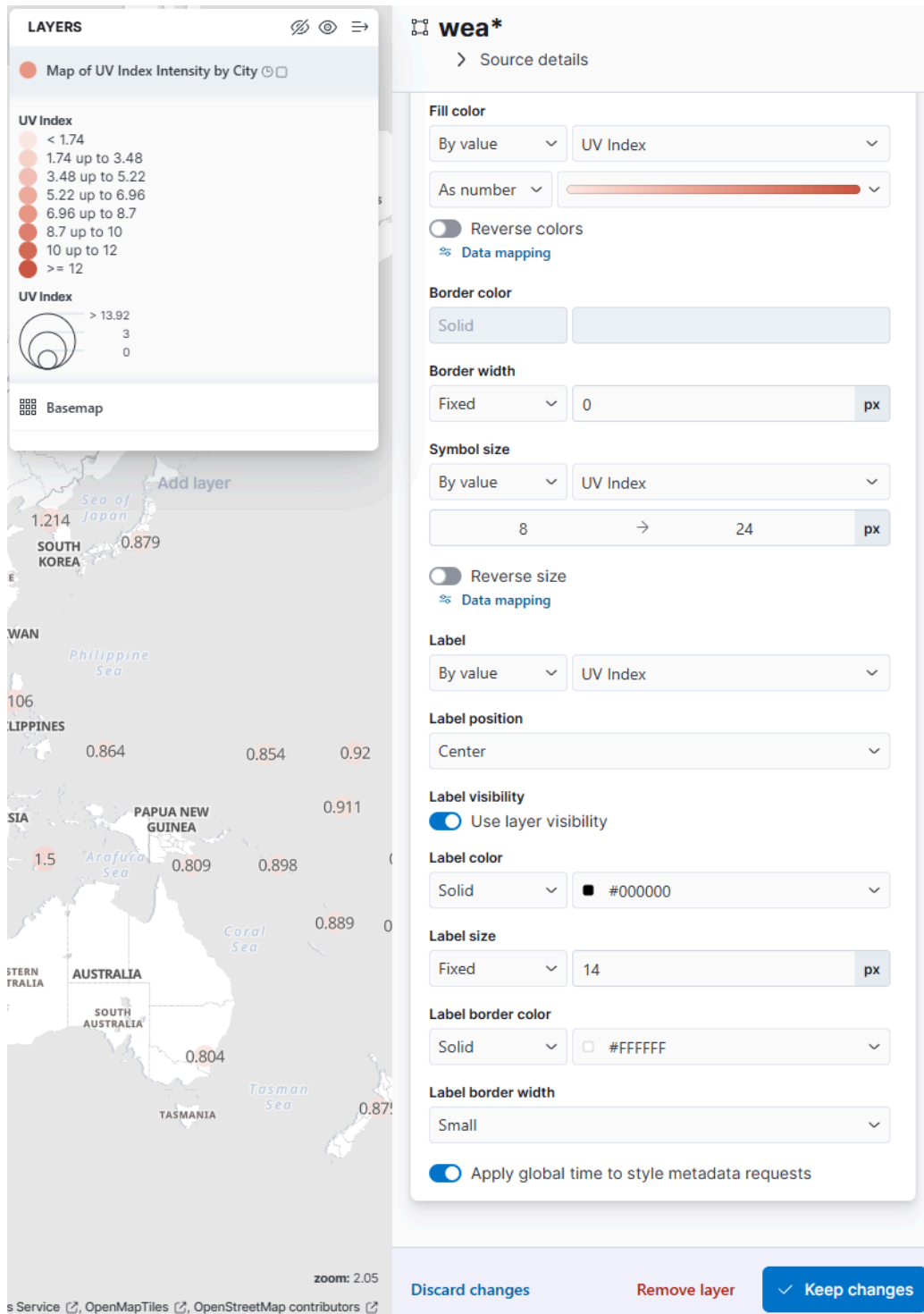
- The Map should then be filled with bubbles (clusters) of different sizes.



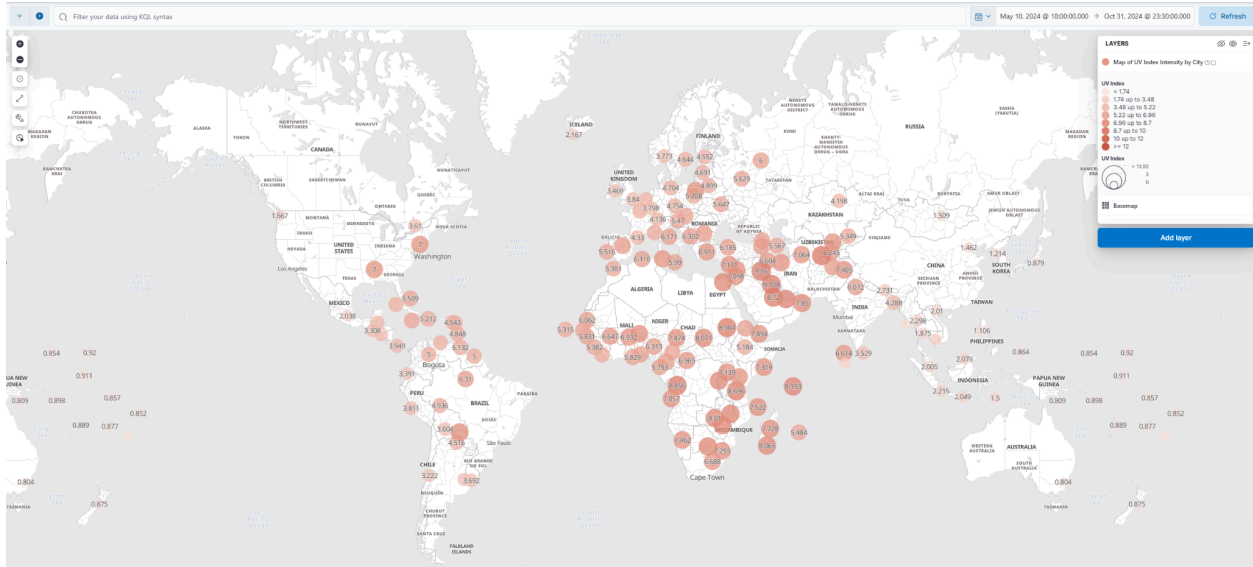
7. In the Metrics section, set the Aggregation field to “Average”, the Field to “uv_index”, and type “UV index” in the Custom Label field.



8. In Layer Style, under Fill color: By Value and UV index.
 - a. As number and, change the color fill to a red gradient (UV low → high). In Symbol size: By value: UV index, making the Min 8 px → Max 24 px
 - b. Click Keep changes



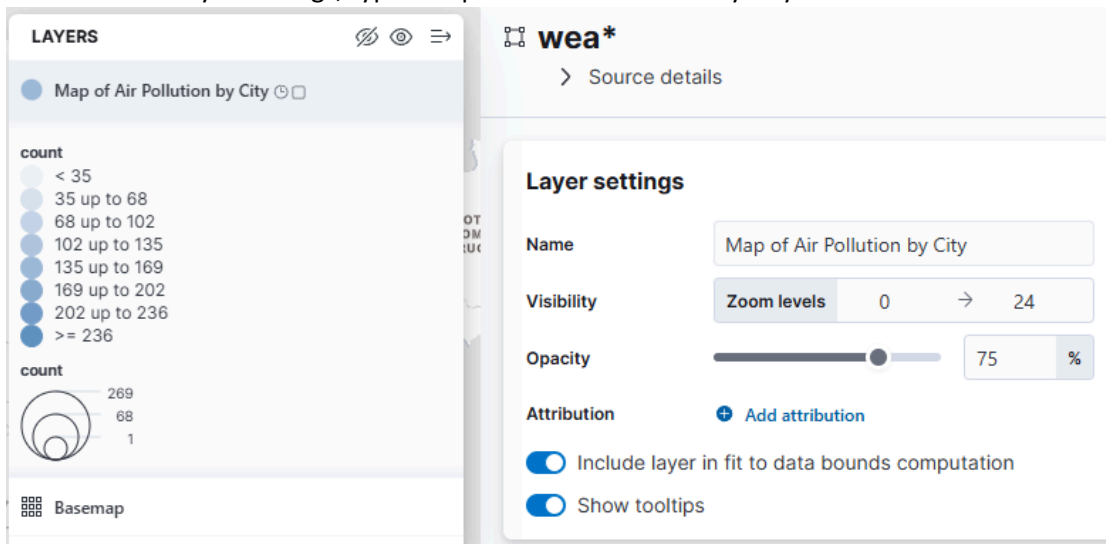
9. The map should display a red gradient of clusters, with darker bubbles indicating high UV Index and lighter bubbles indicating lower UV Index. Save Map.



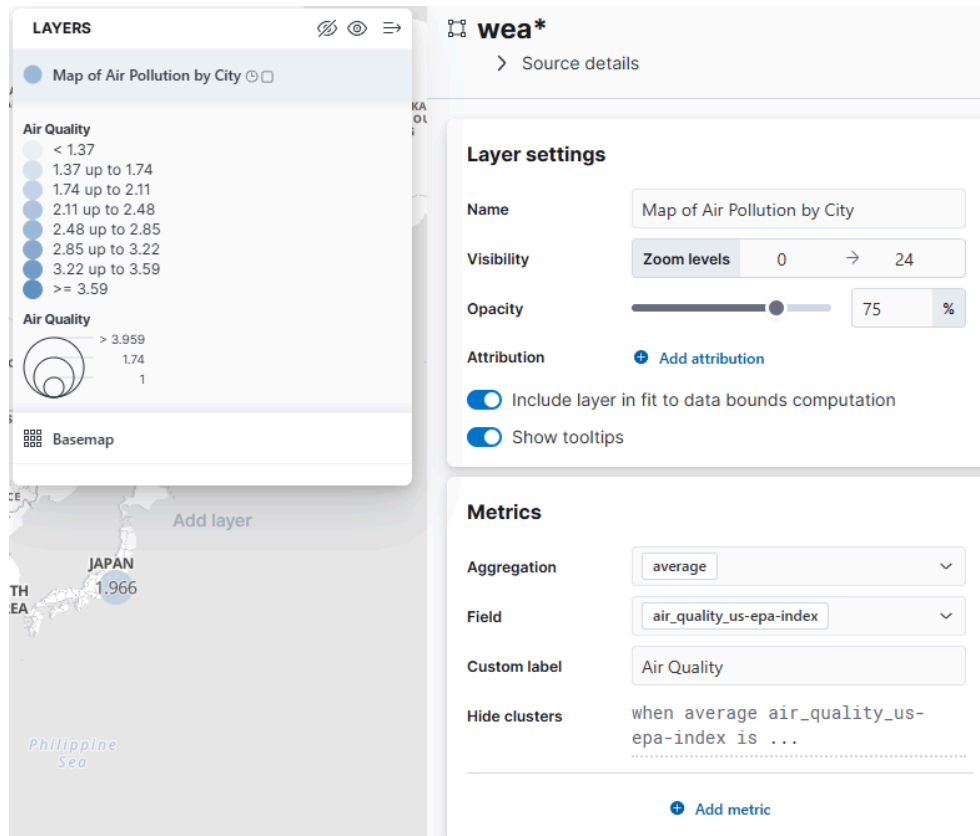
10. Save the Map chart, using the name 'Map of UV Index Intensity by City' and under the section Add to Dashboard, clicking 'none'.

Map- Air Pollution by City

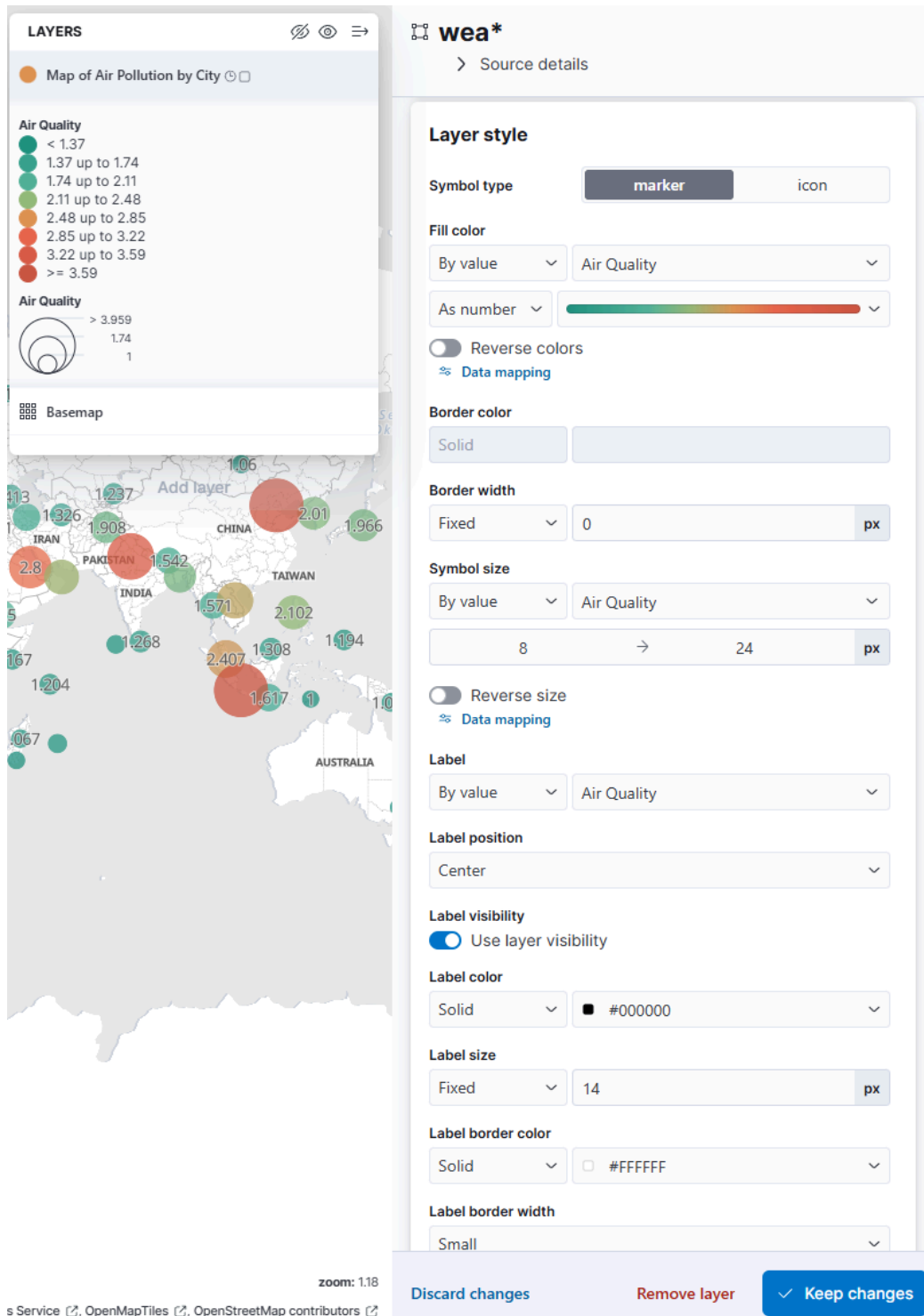
1. Repeat steps 1-3 from the previous map chart, keeping in mind to set the Time Filter.
2. In the Layer settings, type "Map of the Air Pollution by City"



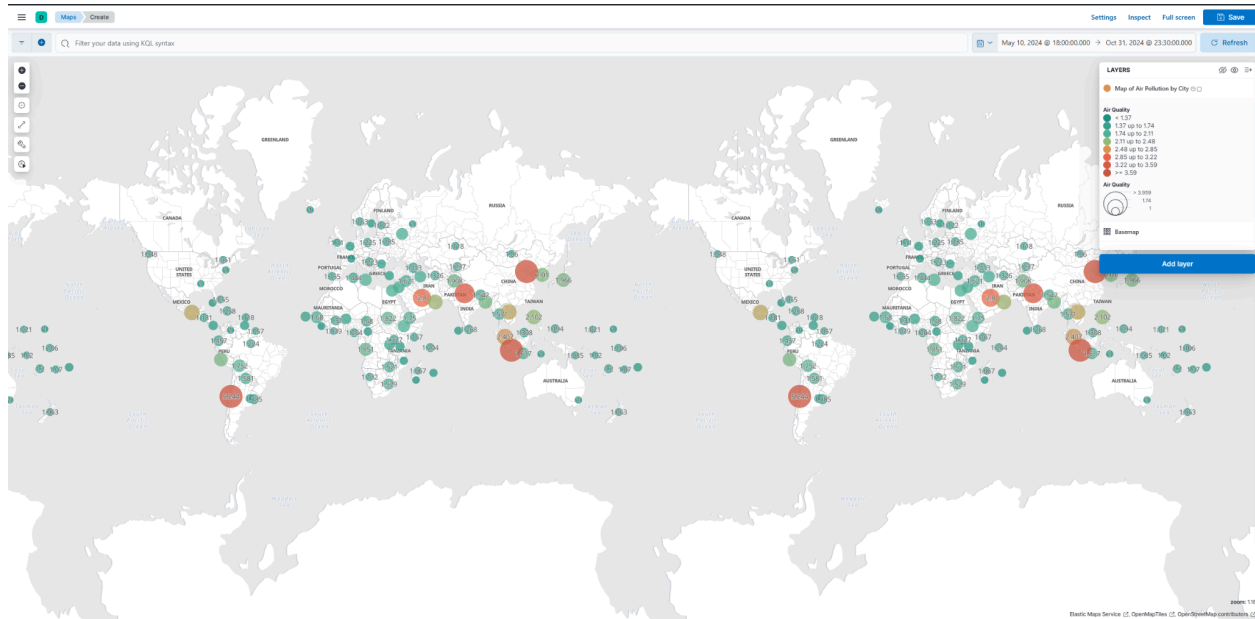
3. In the Metrics section, set the Aggregation field to "Average", the Field to "air_quality_us_epa-index" (Standardized overall air quality index from the US EPA), and type "Air Quality" in the Custom Label field.



4. In Layer Style, under Fill color: By Value and Air Quality.
 - a. As number and, change the color fill to a green-to-red gradient (Green indicating good → red indicating bad)
 - b. Click Keep changes



- The map should produce a green-to-red gradient of clusters that demonstrate air quality. Green bubbles indicate better air quality, while red bubbles indicate poor air quality, according to US EPA Standards. Select Save Map.



6. Save the Map chart, using the name 'Map of the Air Pollution by City' and under the section Add to Dashboard, clicking 'none'.

Dashboard- Global Weather Repository

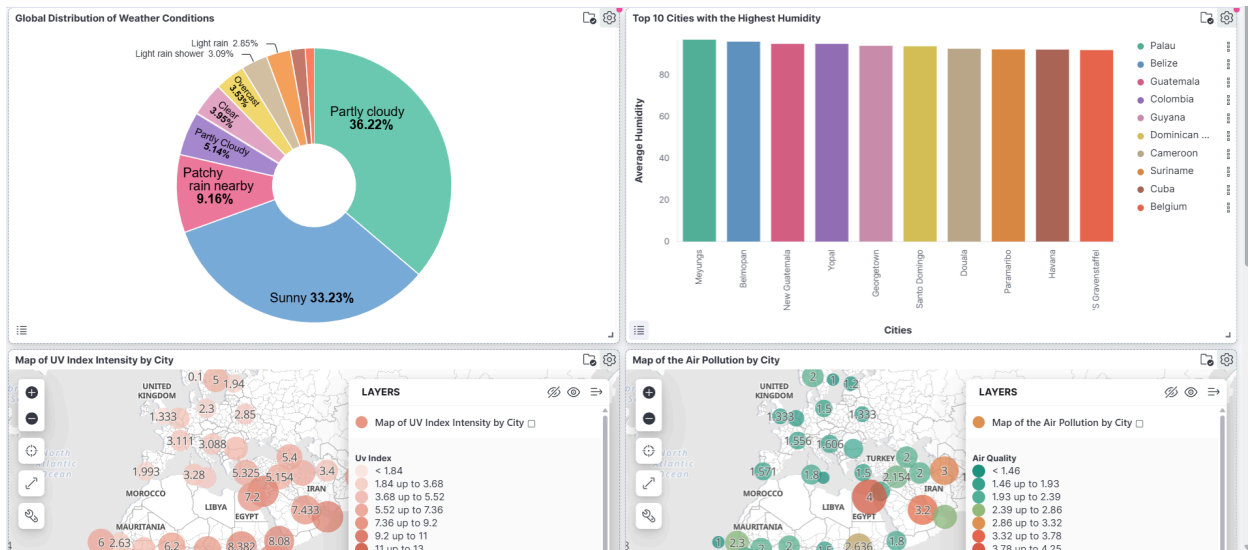
1. Open Kibana: **Analytics** → **Dashboard** → **Create Dashboard**
 - a. On the Dashboard overview page, click Add From Library as the following screenshot:

Add from library

Search...		Types ▾	Tags ▾
T.. ▾	Title	Tags	
	Bar Example		
	Pie Example		
	Markdown Sample		
	Global Distribution of Weather Conditions		
	Top 10 Countries with the Highest Average UV Index		
	Top 10 Cities with the Highest Humidity		

2. Select and add all the charts we have made one by one: Pie, Bar, and Map
 - a. Click the time range in the top-right corner of Kibana. Select Absolute for Start and End Date, set Start Date as May 10, 2024, @ 18:00, and End Date as October 31, 2024, @ 23:30. Then, click update.

3. Select Save and it will open “Save dashboard” as follows
 - a. Name the title of the dashboard as ‘Global Weather Dashboard’. Then, save it.
 - b. The dashboard will look something like this:



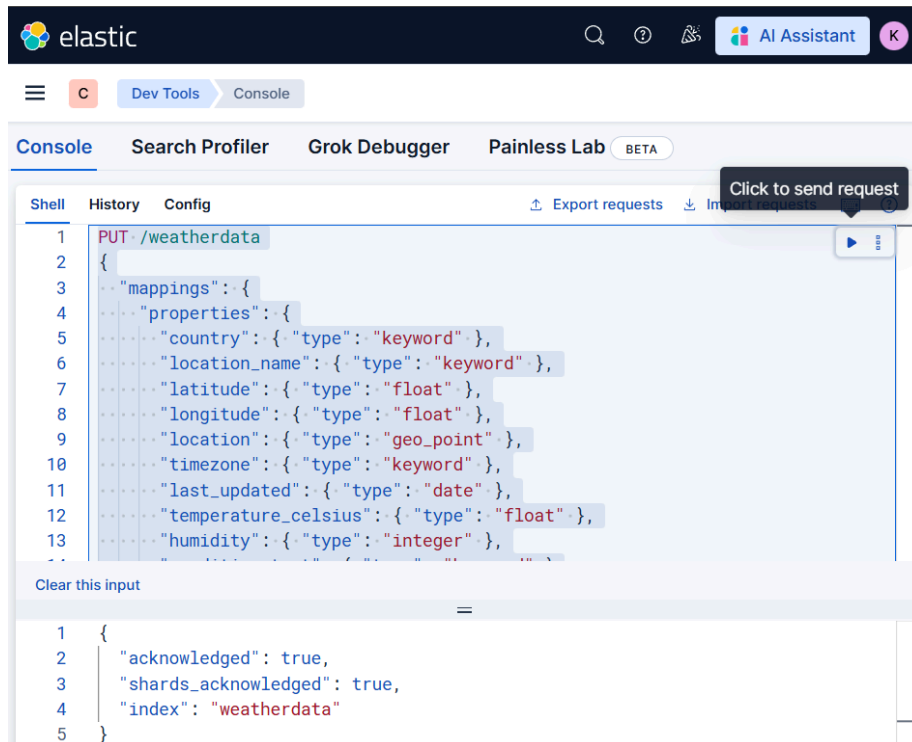
Step 8: ML with Elastic Cloud

This step includes uploading the dataset into Elastic Cloud and using it to create ML algorithms.

1. In Elastic Cloud, go to Management, click Dev Tools, and use this:

```
PUT /weatherdata
```

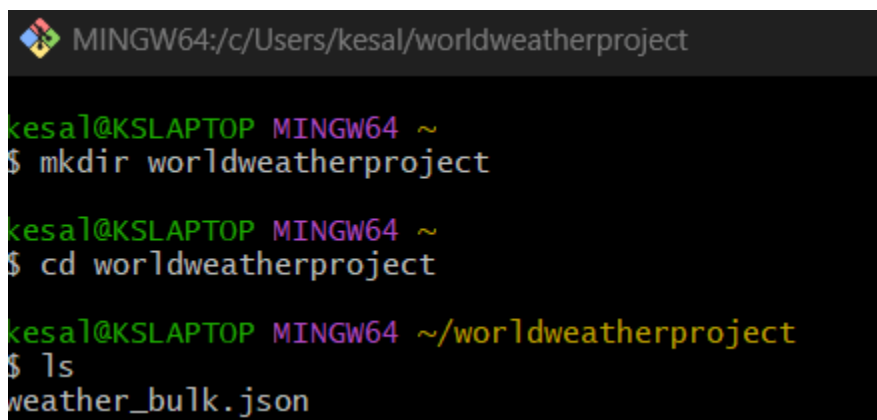
```
{
  "mappings": {
    "properties": {
      "country": { "type": "keyword" },
      "location_name": { "type": "keyword" },
      "latitude": { "type": "float" },
      "longitude": { "type": "float" },
      "location": { "type": "geo_point" },
      "timezone": { "type": "keyword" },
      "last_updated": { "type": "date" },
      "temperature_celsius": { "type": "float" },
      "humidity": { "type": "integer" },
      "condition_text": { "type": "keyword" }
    }
  }
}
```



2. Upload the JSON file to your Elastic Cloud in Gitbash, using this command:

```
cd worldweatherproject
```

```
ls
```



```

curl -H 'Content-Type: application/x-ndjson' \
  -XPOST
  'https://elastic:LChfAbrhEnOPSt6IEuhSDXkl@cis3200-ksalaz63-e75b34.es.
  us-centrall1.gcp.cloud.es.io/weatherdata/_bulk?pretty' \
  --data-binary @weather_bulk.json

```

```
kesal@KSLAPTOP MINGW64 ~/tmp
$ curl -H 'Content-Type: application/x-ndjson' \
  -XPOST 'https://elastic:LChfAbrhEnOPSt6IEuhSDXkl@cis3200-ksalaz63-e75b34.es.us-central1.gcp.cloud.es.io/weatherdata/_bulk?pretty' \
  --data-binary @weather_bulk.json
```

3. In the Dev tool, show the result of the following API console:

GET /_cat/indices?v

GET /_cat/indices?v

is input									
=									
0	0	499b		250b		250b			
green	open								
	temp_celsius_regression								
	e0LDQrLMQh-_pkX4Sk6xag	1	1	355680		18568		73.2mb	
	36.6mb							36.6mb	
green	open								
	weatherdata								
	thynb71tR22cuHLgOD4AKg	1	1	29640		0		28.3mb	
	14.1mb							14.1mb	

Multi-metric Model: multimetric_location_outlier_detector

1. Follow 'Step 5: Create a Data View in Kibana' and 'Step 6: Verify and Update the Geo Field' in the tutorial to maintain consistency in the dataset.
2. In Elastic Cloud, open **Kibana**
 - a. Go to **Analytics** → **Machine Learning** → **Anomaly Detection** → **Create Job** → Click **Wea***.
 - b. Choose Multi-metric
3. In the Multi-metric window, you will have to set up the time range to be May 10, 2024, @ 18:00 (6:00) and October 31, 2024, @ 23:30 (11:30). Clicking next:

Create job: Multi-metric

Using data view weatherbulk

1

Time range

2

Choose fields

3

Job details

4

Validation

5

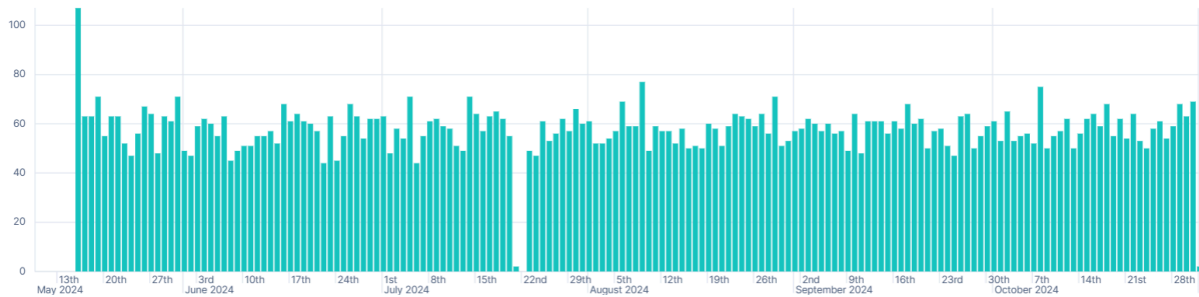
Summary

Time range

May 10, 2024 @ 06:00:00.000 → Oct 31, 2024 @ 23:30:00.000

Use full data

See



Next >

- In the add metrics section, add mean(temperature_celsius), mean(humidity), mean(uv_index), and mean(air_quality_gb-defra-index). For the split field, pick location_name. For influencers, select country, humidity, and UV index. For the bucket span, write 1d. Click Next.

Mean(temperature_celsius)



Mean(humidity)



Mean(uv_index)



Mean(air_quality_gb-defra-index)



Add metric

Split field

A field to split analysis by. Each value of this field will be modeled independently.

location_name

+

Influencers

Categorical fields that influence the results. Who/what might you 'blame' for an anomaly? 1-3 influencers are recommended.

country

humidity

uv_index

+

Bucket span

Sparse data

- In the Job details section, type 'multimetric_location_outlier_detector' in the Job ID and click next.

Create job: Multi-metric

Using data view wea (weatherdata*)



Job details

Job ID

A unique identifier for the job. Spaces and the characters / ? , " < > | * are not allowed

multimetric_location_outlier_detector

Job description

Optional descriptive text.

Groups

Optional grouping for jobs. New groups can be created or picked from the list of existing groups.

Select or create groups

> Additional settings

> Advanced

< Previous

Next >

6. For the Validation, you will see this:

Create job: Multi-metric

Using data view wea (weatherdata*)



Validation

✓ Cardinality

Cardinality of detector fields is within recommended bounds. [Learn more](#)

ⓘ Bucket span

Bucket span is 1 day or more. Be aware that days are considered as UTC days, not local days. [Learn more](#)

✓ Time range

Valid and long enough to model patterns in the data.

✓ Influencer configuration

Passed the validation checks. [Learn more](#)

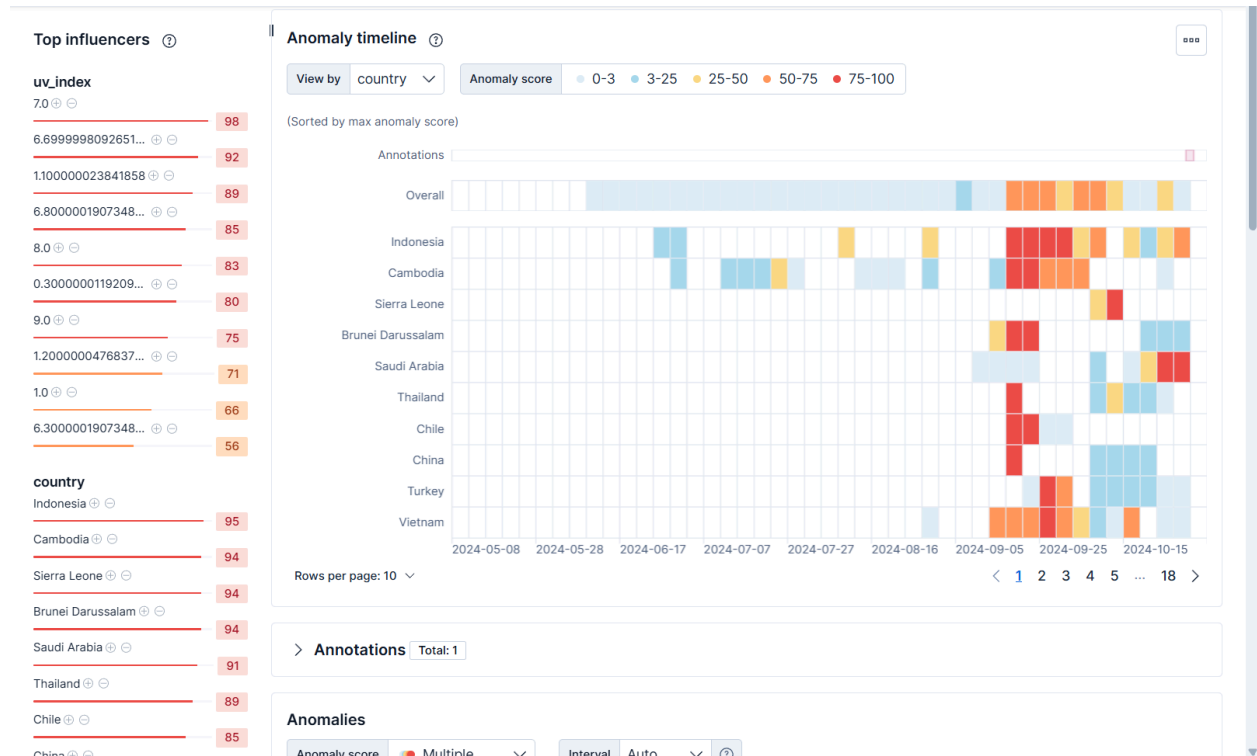
✓ Model memory limit

Valid and within the estimated model memory limit. [Learn more](#)

< Previous

Next >

7. In the summary, click Create job. Once the job is created, click View Results. It will show something like this:



Regression Model- Temperature_Celsius_Regression

1. Select in Kibana: **Analytics** → **Machine Learning** → **Data Frame Analytics** → **Manage Jobs** → **Create Jobs** → **Select wea*** → **Click Regression**. For dependent variables, pick

temperature_celsius.

1 Configuration



Outlier detection

Identify unusual data points in the data set.

Select



Regression

Predict numerical values in the data set.

✓ Selected



Classification

Predict classes of data points in the data set.

Select

Query

Runtime fields

No runtime field

wea

Columns 10/42 Sort fields Histogram charts

air_quality_Carbon_Mono...	air_quality_Nitrogen_diox...	air_quality_Ozone	air_quality_PM10	air_quality_PM2.5	air_quality_Sulphur_dioxi...	air_quality_gb-defra-ind
253.7	0.1	24	9.3	2.2	0.1	
503.2	11.1	78	21.46	21.09	12.58	
353.35	19.24	44	52.91	31.265	11.47	
1,990.6	71.04	38	124.875	121.915	127.28	
244.2	9.065	85	25.16	13.69	10.545	

Rows per page: 5

Dependent variable

temperature_celsius

Included fields

40 fields included in the analysis

2. For the included fields section, make sure to exclude/unselect:

- 1) Air_quality_gb-defra-index
- 2) air_quality_us-epa-index
- 3) condition text
- 4) country,
- 5) feels_like_celsius
- 6) feels_like_fahrenheit
- 7) gust_kph
- 8) last_updated_epoch
- 9) location_name
- 10) moon_phase.keyword
- 11) moonrise.keyword
- 12) moonset.keyword,
- 13) precip_in
- 14) pressure_mb
- 15) temperature_fahrenheit
- 16) timezone
- 17) visibility_km
- 18) sunrise.keyword
- 19) sunset.keyword
- 20) wind_direction.keyword,
- and
- 21) wind_mph.

Included fields

19 fields included in the analysis

Is included

Is not included

☐	Field name	Mapping	Is included	Is required	Reason
☒	 air_quality_Carbon_Mo noxide	float	Yes	No	
☒	 air_quality_Nitrogen_di oxide	float	Yes	No	
☒	 air_quality_Ozone	float	Yes	No	
☒	 air_quality_PM10	float	Yes	No	
☒	 air_quality_PM2.5	float	Yes	No	

Rows per page: 5

<

1

2

3

4

5

...

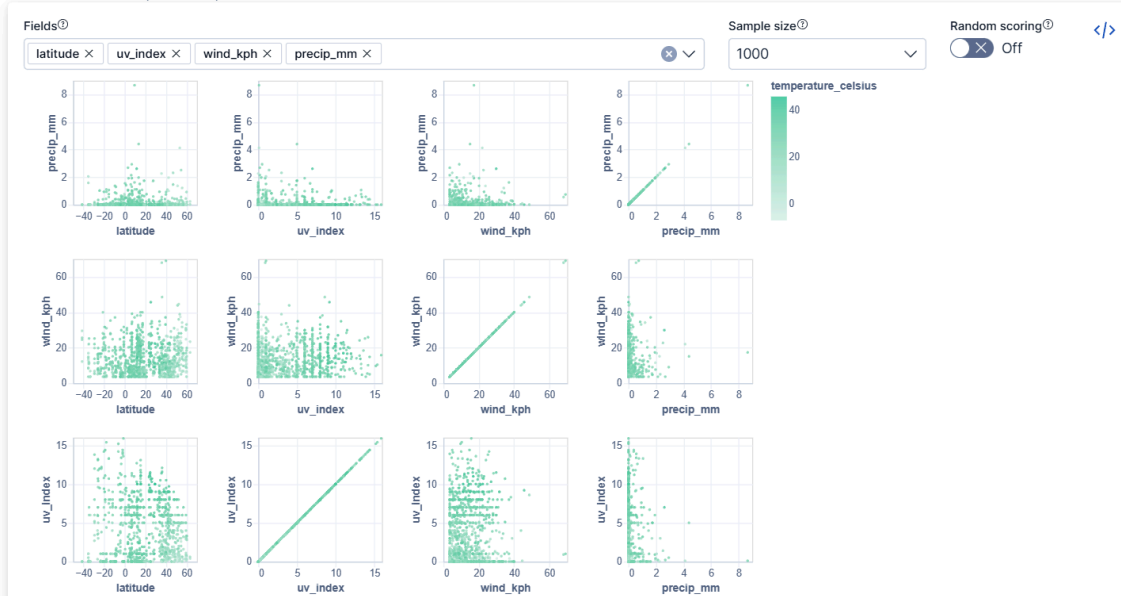
10

>

3. In the scatterplot matrix, add the fields latitude, uv_index, wind_kph, and precip_mm.

Scatterplot matrix

Visualizes the relationships between pairs of selected included fields.



- Maintaining a training percentage of 80% and clicking next.
- In the additional options section, set it to 5. Set the model memory limit to 130mb. Click Continue.

2

Additional options

Advanced configuration

Feature importance values

5

Specify the maximum number of feature importance values per document to return.

Prediction field name

Define the name of the prediction field in the results. The default is <dependent_variable>_prediction.

Randomize seed

The seed for the random generator used to pick training data.

Model memory limit

130mb

☒ Use estimated model memory limit

The approximate maximum amount of memory resources that are permitted for analytical processing.

Maximum number of threads

1

The maximum number of threads to be used by the analysis. The default value is 1.

- In the Job detail section, type 'temperature_celsius_regression' and click Next. For validation, it will appear as follows, and you will continue to click Create.

3 Job details

Job ID	Job description	Destination index
tempe_celsius_regression		tempe_celsius_regression

[Edit](#)

4 Validation

✓ Dependent variable

The dependent variable field contains continuous values suitable for regression analysis.

✓ Training percent

The training percent is high enough to model patterns in the data.

⚠ Feature importance

Enabling feature importance can result in long running jobs when there are a large number of training documents.

✓ Analysis fields

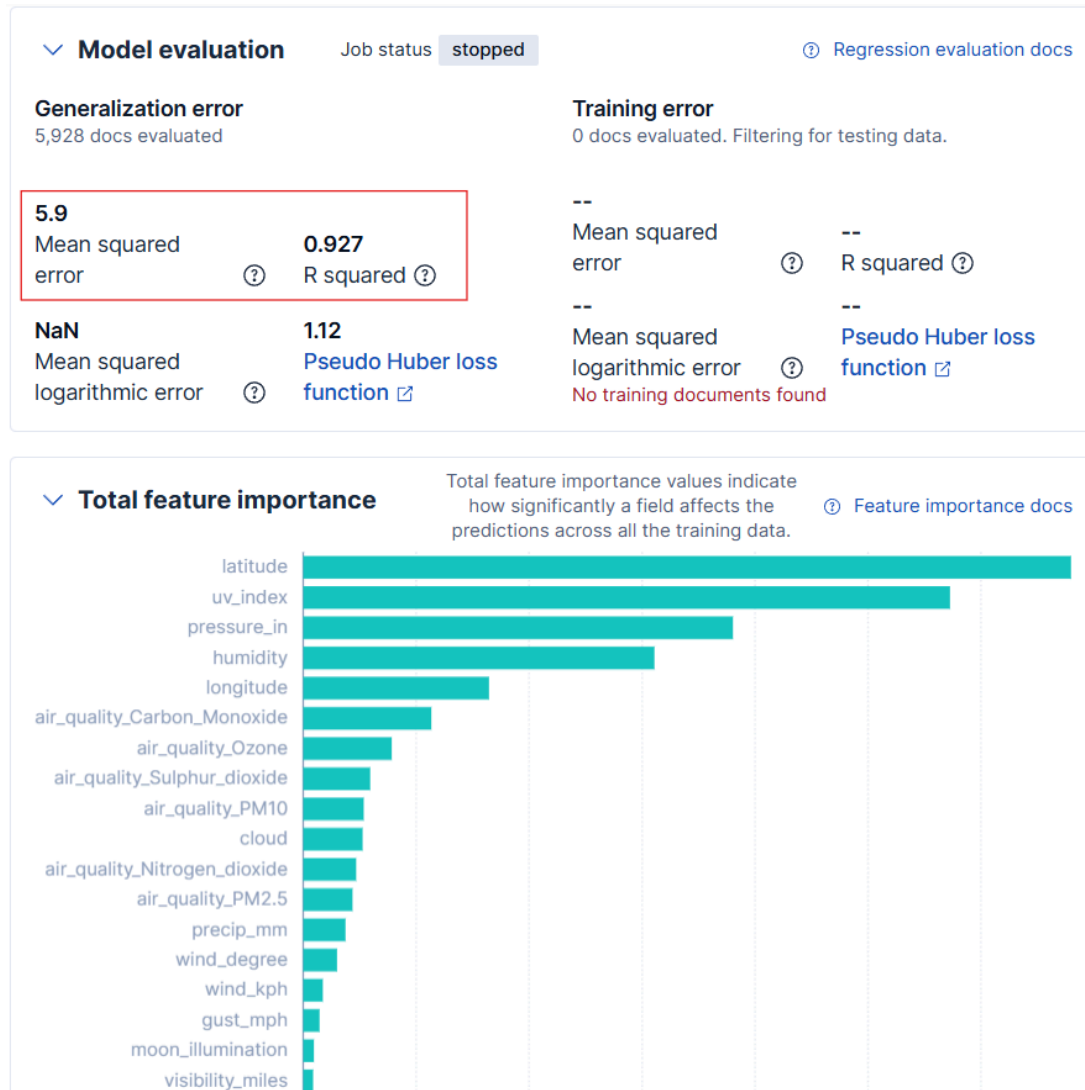
The selected analysis fields are at least 70% populated.

[Continue](#)

7. You will see the page with Data Frame Analytics. Select Data Frame Analytics. You will then need to wait for the synchronization of saved objects after you synchronize them. You will then be able to open it up when it finishes all its phases.

ID	Description	  ☐☐☐ >
tempe_celsius_regression		
Memory status	Source index	
ok	weatherdata*	
Destination index	Type	
tempe_celsius_regression	regression	
Status	Progress	
started	Phase 4/8	
Spaces		
C		

8. You can see the accuracy of the model with the testing data set. Select Testing, and it will show the following:



References

1. URL of Data Source:

<https://www.kaggle.com/datasets/nelgiriyeewithana/global-weather-repository>

To get the JSON file (weather_bulk.json), use the GitHub link

2. URL of your GitHub:

<https://github.com/ksalaz63/GlobalWeatherAnalysis>

3. URL of References:

<https://www.elastic.co/guide/en/kibana/current/reporting-getting-started.html>